Static Analysis of Cable-Driven Manipulators With Non-Negligible Cable Mass

Kris Kozak, Qian Zhou, and Jinsong Wang

***Abstract*—This paper addresses the static analysis of cable-driven robotic manipulators with non-negligible cable mass. An approach to computing the static displacement of a homogeneous elastic cable is presented. The resulting cable-displacement expression is used to solve the inverse kinematics of general cable-driven robotic manipulators. In addition, the sag-induced stiffness of the cables is derived. Finally, two sample robotic manipulators with dimensions and system parameters similar to a large scale cable-driven manipulator currently under development are analyzed. The results show that cable sag can have a significant effect on both the inverse kinematics and stiffness of such manipulators.**

***Index Terms*—Cable-driven robotic manipulators, kinematics, modeling, statics, stiffness.**

I. Introduction

IN THIS PAPER, the effect of cable mass in the static analysis of cable-driven robotic manipulators is analyzed. Cables with significant mass will tend to sag under their own weight. Deviation from an idealized massless, straight cable poses significant challenges for an otherwise simple and straightforward analysis. The inverse kinematics of cable-driven manipulators, for example, lose their closed-form solvability when cable mass is non-negligible. Furthermore, cable sag affects numerous performance characteristics of cable-driven manipulators, including usable workspace, positioning accuracy, and stiffness.

A. Motivation

Although many cable-driven robotic manipulators are assumed to have negligible cable mass, the problem of analyzing and understanding the effects of cable sag is relevant to almost all manipulators. Minimally, analytical tools are required to determine the consequences of assuming cable mass to be negligible. When cable mass is non-negligible, a comprehensive understanding of cable statics provides a basis for good design and analysis. Thus, the research presented in this paper is particularly applicable to large-scale and/or underconstrained manipulators, which are more likely than other types of manipulators to have non-negligible cable mass.

The motivation for this research comes directly from challenges in the design and control of a proposed large-scale, cable-driven robotic manipulator: the receiver (feed) support system of a large radio telescope named the Five-Hundred Meter Aperture Spherical Radio Telescope (FAST) [1]–[3]. Fig. 1 shows a conceptualization of the FAST system, where the receiver cabin is suspended by cables over a large reflector dish. The challenge for FAST viability is ultimately one of precision control. To observe 5 GHz radio waves, the astronomical receiver is required to follow trajectories along a focal surface with a precision of 4 mm and 0.1° root mean square (RMS). The feed support system of FAST consists of two layers: the receiver cabin positioning system, which is simply a large-scale, cable-driven manipulator that provides the coarse positioning, and a fine-tuning manipulator (Gough–Stewart platform) that can compensate for the coarse positioning errors and achieve the desired precision. However, due to various limitations in sensors and other hardware, the motion of the system must be predicted relatively accurately over short intervals of time to accurately compensate for errors in positioning. This means that the characteristics of the cables must be well understood, not only for the problem of design, but also as a basis for performing accurate model-based prediction.

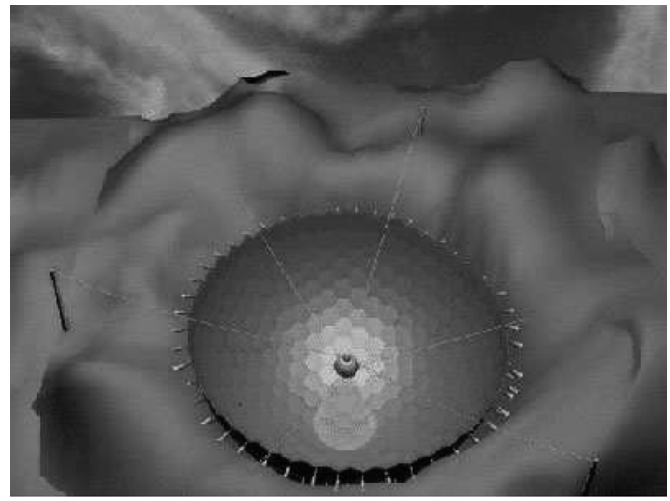

Fig. 1. Conceptual model of the FAST system.

B. Issues Addressed

This paper addresses two main issues related to cable-driven robotic manipulators: computation of the inverse kinematics, and the computation of the stiffness of a manipulator due to the sag in its cables. The process for solving these problems begins from the basic analysis of a single cable, which, in turn, is based on the static displacement or shape of that cable. The static displacement solution can be used to obtain the inverse

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kinematics, the stiffness of a single cable, and the stiffness of an entire robotic manipulator. These results can then be used to fully analyze the statics of that manipulator.

C. Applications

This paper presents basic analytical tools for studying large-scale cable-driven robotic manipulators, so the results are applicable to a variety of problems. This work can be most directly applied to the static and dynamic modeling of manipulators with significant cable sag. Furthermore, the tools developed for computing the inverse kinematics and stiffness are directly applicable to the manipulator design problem of workspace optimization. Similarly, this work provides a basis for analyzing manipulators based on stiffness, and the work can be extended to include well-established Jacobian analysis. Finally, the inverse kinematics and stiffness derivations are applicable to real-time control and controller design.

D. Related Work

A significant amount of work has been done in both the areas of cable-driven robotic manipulators and sagging cables. With regard to cable-driven manipulators, research has been performed on everything from practical implementations, such as the fairly large-scale NIST ROBOCRANE [4], and the high-speed FALCON [5] and WARP [6] manipulators, to the design of the large-scale FAST system [2] (which includes some cable-sag results), to numerous studies on kinematics and workspace [6]–[8], manipulability [9], [10], and traditional stiffness analyses [11]. Most of this research, however, does not provide much direct insight into how cable sag affects these manipulators.

Research is also plentiful in the area of cable sag and cable mechanics. Many of the problems and solutions of cable mechanics related to this paper are discussed by Irvine in his book *Cable Structures* [12]. Beyond the basics, much of the current research in cable sag relates to bridges and guys, and addresses topics such as dynamic modeling and vibration control [13], [14]. Ultimately, though, none of the previous research quite adequately addresses the specific issues of modeling and analyzing cable sag for cable-driven manipulators; this paper seeks to fill that gap.

E. Assumptions

In this paper, we restrict our analysis to cable-driven robotic manipulators with the following characteristics. The manipulator is static or moving slowly enough to be considered static. Gravity is non-negligible. The cables have homogeneous properties and negligible flexural stiffness. Unless stated otherwise, the cables are assumed to be elastic. Furthermore, the drawing points of the cables are assumed to be fixed.

II. Computing the Static Cable Profile

The first challenge for analyzing a cable-driven robot with sagging cables is to mathematically describe the static displacement (or the shape) of the cables under the influence of gravity. To compute the static displacement of a cable, we will assume

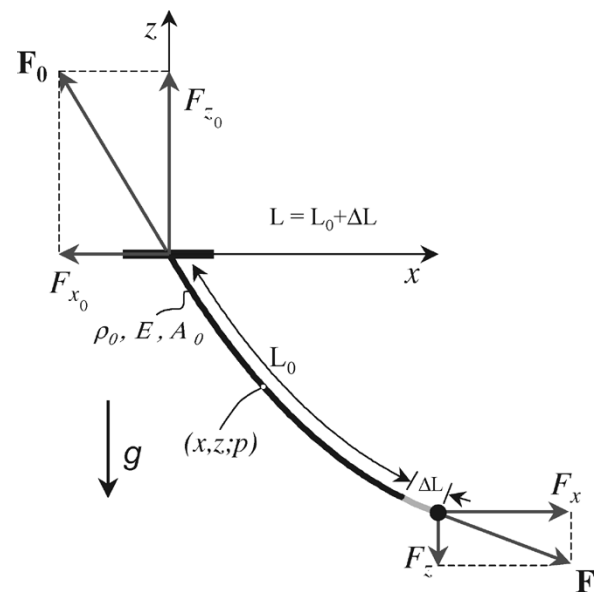

Fig. 2. Diagram of a sagging cable.

for the moment that the unstrained length of the cable, L_0, and the force applied to the unfixed end of the cable, $\mathbf{F}$, are known. Fig. 2 shows the relevant coordinates and parameters of the cable system, where ρ_0, E, A_0, and ΔL represent the unstrained linear density, the elastic modulus, the unstrained cross-sectional area, and the strain of the cable, respectively.

As shown in the figure, a point along the length of the strained cable can be denoted by Cartesian coordinates x and z, and the path variable p. The variable p represents the strained length of the cable segment as measured from the attachment point of the cable to the point (x, z). The variable s will be used to denote the unstrained length of that cable segment. From this starting point, the equations for the shape of the cable can be derived.

A. Derivation

Irvine notes that the solution for the displacement of a non-symmetric sagging elastic cable has been known since at least the 1930s [12], and thus there is little necessity to derive it again from scratch here. However, as several components of the cable profile derivation are relevant to the work presented in this paper, we will briefly reproduce Irvine's derivation, using our coordinate system and variables.

To begin with, the shape of the cable must satisfy the geometric constraint

$$\left(\frac{dx}{dp}\right)^2 + \left(\frac{dz}{dp}\right)^2 = 1. \tag{1}$$

At point p, the force balance for the segment of the cable anchored to ground can be written as

$$T\frac{dx}{dp} = F_x \tag{2}$$

$$T\frac{dz}{dp} = F_z + \rho_0 g(s - L_0) \tag{3}$$

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where T is the tension in the cable at point p. Solving for dx/dp and dz/dp, and substituting into (1), we find an expression for the tension as a function of the unstrained path length s

$$T(s) = \sqrt{F_x^2 + (F_z + \rho_0 g(s - L_0))^2}. \tag{4}$$

The tension must satisfy Hooke's law

$$T(s) = EA_0 \left(\frac{dp}{ds} - 1 \right). \tag{5}$$

Substituting the tension expression (4) back into (2) and (3), and noting that $dx/ds = dx/dp\, dp/ds$ and $dz/dx = dz/dp\, dp/ds$, the following solvable differential equations are obtained:

$$\frac{dx}{ds} = \frac{F_x}{EA_0} + \frac{F_x}{\sqrt{F_x^2 + (F_z + \rho_0 g(s - L_0))^2}} \tag{6}$$

$$\frac{dz}{ds} = \frac{F_z}{EA_0} + \frac{\rho_0 g(s - L_0)}{EA_0} + \frac{F_z + \rho_0 g(s - L_0)}{\sqrt{F_x^2 + (F_z + \rho_0 g(s - L_0))^2}}. \tag{7}$$

Integrating and applying the boundary conditions $x(0) = 0$ and $z(0) = 0$, we get

$$x(s) = \frac{F_x s}{EA_0} + \frac{|F_x|}{\rho_0 g} \left[\sinh^{-1} \left(\frac{F_z + \rho_0 g(s - L_0)}{F_x} \right) + - \sinh^{-1} \left(\frac{F_z - \rho_0 g L_0}{F_x} \right) \right] \tag{8}$$

$$z(s) = \frac{F_z}{EA_0} s + \frac{\rho_0 g}{EA_0} \left(\frac{s^2}{2} - L_0 s \right) + \frac{1}{\rho_0 g} \left[\sqrt{F_x^2 + (F_z + \rho_0 g(s - L_0))^2} - \sqrt{F_x^2 + (F_z - \rho_0 g L_0)^2} \right]. \tag{9}$$

Here we note that an absolute value appears in the $x(s)$ solution to ensure that the cable profile is valid for negative values of F_x; this point is omitted by Irvine. In theory, the static profile of any homogeneous, elastic cable under the influence of gravity, and acted on by a force applied to the end of the cable, can be solved for using these equations. The resulting curve is generally referred to as an elastic catenary.

1) Inextensible Cables: For some situations, the elasticity of the cable can be neglected, and the solution can be reduced to the standard catenary. Inextensibility requires infinite stiffness, which can be introduced by taking the limit as the elastic modulus E approaches infinity. Among other things, this simplification allows the cable profile to be written as an explicit function of x, and the length to be written as an explicit function of the applied forces and cable endpoint position.

B. Experimental Results

In this section, the theoretical cable shape, given by (8) and (9), is verified experimentally. Although the solution of a sagging cable is relatively well known, it is difficult to find experimental verification of the purely static characteristics of the catenary. However, Russell and Lardner investigated the statics of an elastic cable, finding a persistent 2.5% difference between the model and their experimental results [15]. Here, we experimentally investigate the shape of a sagging cable. For the purposes of this paper, this shape verification is performed primarily as a setup for the stiffness experimentation done in Section IV-B.

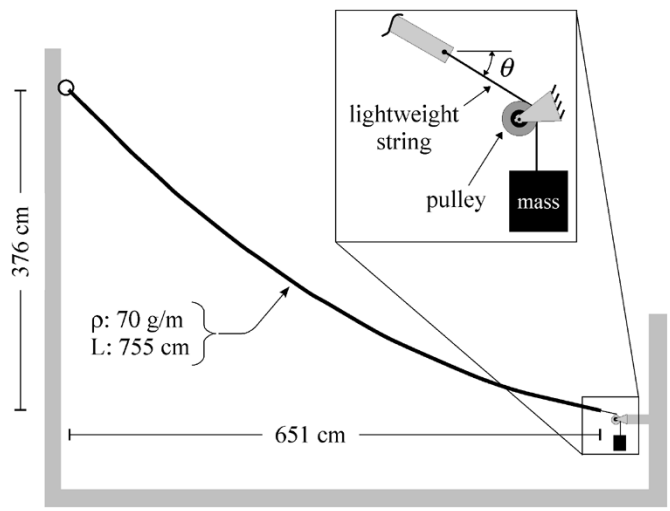

Fig. 3. Experiment setup.

TABLE I
ROPE PROPERTIES

Rope and Experiment Properties	
Rope Length	755 cm
Diameter	10.5 mm
Density	70 g/m
Rated Elongation (80kg load)	7.8%
Nominal Load Mass	623g

The setup chosen for measuring cable sag is shown in Fig. 3. The setup can be described as follows: a length of dynamic climbing rope (the sort used in mountain climbing) 755 cm in length was fixed at one end; this attachment point was designated as the origin of the coordinate system. The other end of the rope was attached to a lightweight string that was fed through a pulley and attached to a weight with a mass of 623 g to produce a nominal force with a known magnitude of approximately 6.1 N. The climbing rope, with properties listed in Table I, was used for this experiment because it has a relatively high density, relatively low bending stiffness, and because its properties are sufficiently homogeneous along its length. Most other commonly available kinds of ropes and cables lack at least one of these characteristics for lengths of less than 10 m. The linear density of the rope was listed as 70 g/m by the manufacturer and was confirmed by an independent measurement. The nominal endpoint position of the rope was measured to be $(x, y) = (651 \text{ cm}, -376 \text{ cm})$.

The experiment was performed indoors to avoid wind and other disturbances. After the cable was in place, its shape was measured. The position of the cable was measured directly using a simple plumb-line measurement technique at multiple points

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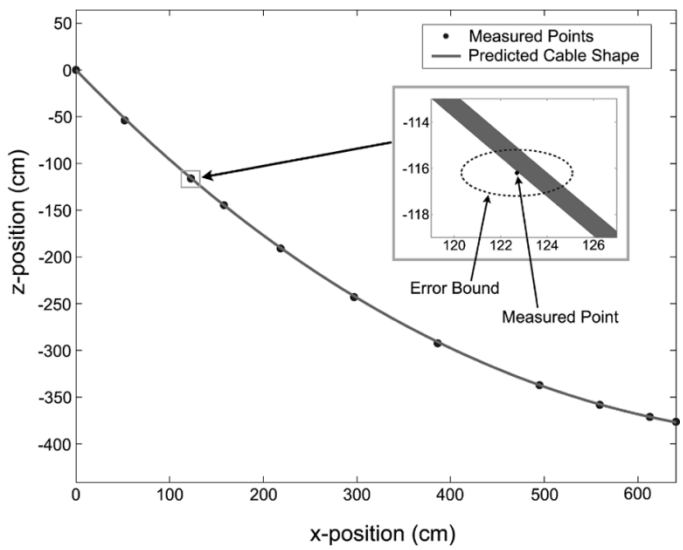

Fig. 4. Measured and predicted cable shape.

along its length. The predicted shape of the cable was computed based on the applied force. The magnitude of the applied force was known from the weight, and the direction of the force was measured by analyzing close-up digital images of the region around the endpoint of the cable, obtained using a digital camera. The mean value of these measurements was used to generate the prediction. The accuracy of the plumb-line measurements varied from approximately 3 cm horizontal and 1 cm vertical at the high, fixed end of the cable, to approximately 1 cm horizontal and 1 cm vertical at the applied-force end. The digital-image-based measurements of the cable slope have an estimated error of less than 0.004 rad (0.23°), which is based on the resolution of the camera and the observed variance in the measurements.

Fig. 4 shows the measured and predicted shapes of the rope. The agreement between the predicted shape and the measured shape appears very close at the provided level of zoom, and indeed the agreement is quite good. The inset shows a close-up of one of the measured points and its estimated measurement error in relationship to the predicted position of the rope. In fact, all of the points measured vary from the predicted position of the rope by less than the estimated error bounds. It should be noted that the elasticity of the cable was relatively insignificant. In conclusion, to the level of precision attained by the measurements, this experiment confirmed that the predicted cable shape, given by (8) and (9), describes the physical system accurately.

III. Inverse Kinematics

When cable sag is non-negligible, the inverse kinematics of a cable-driven robot become quite difficult to compute. The inverse kinematics must be solved as a statics problem rather than as a pure kinematics problem. In this section, the inverse kinematics problem is posed, and a solution method is presented. Inverse kinematics solutions must be divided into three different types, based upon the degree of constraint of the manipulator.

A. Minimally Constrained Systems

The most straightforward inverse kinematics problem is that of the minimally constrained system, where the number of cables is the same as the number of degrees of freedom (DOFs). Gravity plays the role of the $n+1$th cable that is needed to fully constrain the system [5], [9]. The equations for this type of system consist of a set of statics equations and a set of kinematic-static equations. The position and orientation of the end-effector are specified. The unknown variables are the cable lengths and the applied forces. For both planar and spatial manipulators, the cable forces can be parameterized by two variables—each cable force vector is constrained to lie in the same vertical plane as the imaginary line connecting the drawing point to the attachment point of the cable. Thus, for a manipulator with n cables, there will be a total of $3n$ unknowns. The equations for static equilibrium are

$$\sum_{i=1}^{n} \mathbf{R}_i(-\mathbf{F}_i) + m\mathbf{g} = \mathbf{0} \tag{10}$$

$$\sum_{i=1}^{n} \mathbf{r}_i \times \mathbf{R}_i\mathbf{F}_i = \mathbf{0} \tag{11}$$

where each $\mathbf{F}_i$ is the vector quantity of the ith cable force (like $\mathbf{F}$ in Fig. 2). $\mathbf{R}_i$ is the rotation matrix that maps the force vector $\mathbf{F}_i$ from the cable frame to the fixed/manipulator frame. The vector $\mathbf{r}_i$ is the vector from the center of mass of the mobile platform to the attachment point of cable i. Each cable force $\mathbf{F}_i$ consists of two unknown components, F_{xi} and F_{zi}.

The necessary kinematic-static equations come directly from (8) and (9). The endpoint of the ith cable is given by

$$x_{i\mathrm{end}} = x_i(L_{0i}) = \frac{F_{xi}L_{0i}}{EA_0} + \frac{|F_{xi}|}{\rho_0 g}\left[\sinh^{-1}\left(\frac{F_{zi}}{F_{xi}}\right) - \sinh^{-1}\left(\frac{F_{zi}-\rho_0 g L_{0i}}{F_{xi}}\right)\right] \tag{12}$$

$$z_{i\mathrm{end}} = z_i(L_{0i}) = \frac{F_{zi}L_{0i}}{EA_0} - \frac{\rho_0 g L_{0i}^2}{2EA_0} + \frac{1}{\rho_0 g}\left[\sqrt{F_{xi}^2 + F_{zi}^2} - \sqrt{F_{xi}^2 + (F_{zi} - \rho_0 g L_{0i})^2}\right]. \tag{13}$$

Because both $x_{i\mathrm{end}}$ and $z_{i\mathrm{end}}$ are known when the position and orientation of the platform are specified, these equations (one for each of the n cables) can be solved simultaneously with (10) and (11) to find the unknown forces and unstrained cable lengths. Except for some special cases, these equations must be solved numerically.

Obtaining the appropriate length of the cables is only part of the actual inverse kinematics problem; the other part is controlling the length precisely. This may require finding a relationship between the cable length and the angular position of the reeling mechanism. Unfortunately, without knowing the specifics of the cable-reeling mechanism, we cannot determine beforehand whether the angular position of the reeling mechanism will have

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a direct relationship to the unstrained cable length. In some cases, the strained length may be more appropriate.

From Hooke's law in (5), the strained path length p can be integrated out as a function of the unstrained path length s (omitted here due to length). Evaluating at $s = L_0$ and simplifying, the following expression for the strained cable length is retrieved:

$$\begin{aligned} L_i = L_{0i} &+ \frac{1}{2\rho_0 g E A_0} \\ &\times \left[F_{zi}\sqrt{F_{xi}^2 + F_{zi}^2} + F_{xi}^2 \sinh^{-1}\left(\frac{F_{zi}}{|F_{xi}|}\right) + \ldots \right. \\ &\quad - (F_{zi} - \rho_0 g L_{0i})\sqrt{F_{xi}^2 + (F_{zi} - \rho_0 g L_{0i})^2} + \ldots \\ &\quad \left. - F_{xi}^2 \sinh^{-1}\left(\frac{F_{zi} - \rho_0 g L_{0i}}{|F_{xi}|}\right)\right]. \end{aligned} \tag{14}$$

In all likelihood, the required reel position will not be easy to predict, so the inverse kinematics solution may be better suited for design and simulation purposes than for real-world open-loop control of the system. Nevertheless, the inverse kinematics solution presented here provides a basis for further studying and analyzing cable-driven robotic manipulators.

1) Inextensible Cable Simplification: For manipulators where cable elasticity is negligible, the inverse kinematics problem can be simplified. By taking the limit of (12) as the Young's modulus, E, approaches infinity, and $x_{i\mathrm{end}}$ becomes

$$x_{i\mathrm{end}} = \frac{|F_{xi}|}{\rho g}\left[\sinh^{-1}\left(\frac{z_i}{F_{xi}}\right) - \sinh^{-1}\left(\frac{F_{zi} - \rho g L_i}{F_{xi}}\right)\right]. \tag{15}$$

Solving for L_i, we get

$$L_i = -\frac{1}{\rho g}\left[F_{xi}\sinh\left(\sinh^{-1}\left(\frac{F_{zi}}{F_{xi}}\right) - \frac{\rho g}{|F_{xi}|}x_{i\mathrm{end}}\right) - F_{zi}\right]. \tag{16}$$

Substituting each L_i into (13), taking the limit as E approaches infinity, and simplifying yields the following equation for $z_{i\mathrm{end}}$:

$$\begin{aligned} z_{i\mathrm{end}} = \frac{|F_{xi}|}{\rho g} &\left[\sqrt{\left(\frac{F_{zi}}{F_{xi}}\right)^2 + 1}\right. \\ &\left. + \ldots - \cosh\left(\sinh^{-1}\left(\frac{F_{zi}}{F_{xi}}\right) - \frac{\rho g}{|F_{xi}|}x_{i\mathrm{end}}\right)\right]. \end{aligned} \tag{17}$$

This equation can now be solved simultaneously with (10) and (11) to find the unknown forces. The forces can then be substituted back into (16) to find the cable lengths and complete the inverse kinematics solution. This simplification allows the forces to be solved independently from the lengths, thus reducing the number of variables that must be solved numerically by n, the number of cables.

B. Underconstrained and Overconstrained Systems

The inverse kinematics solution method for minimally constrained systems also provides a basis from which to solve both underconstrained and overconstrained systems.

1) Underconstrained Systems: Underconstrained systems are those with fewer cables than DOFs. Inverse kinematics for underconstrained systems is a two-part problem. The first part is to choose an attainable or valid end-effector configuration from which to calculate the inverse kinematics. Like reduced mobility or restricted DOF parallel manipulators, underconstrained cable-driven manipulators cannot be positioned arbitrarily in six DOFs.[1]

The second part of the problem is to compute the inverse kinematics for a validly chosen end-effector configuration. Although this paper does not address the first part of the problem, the solution to the latter part can be used to verify whether an end-effector configuration is valid or not. Assuming that a point is in the workspace of the manipulator (and thereby valid), (10)–(13) will also be valid. These equations are solved as before, but as there are fewer unknown variables, fewer equations are required to generate a solution. After a solution is found, the results can be substituted back into the remaining equations to verify whether the specified end-effector configuration is valid (contradictions in the results after substitution imply that the specified end-effector configuration is not valid).

2) Overconstrained Systems: Overconstrained systems can also be solved readily using the techniques from the previous section. Equation (17) must again be solved numerically, but the unknown variables outnumber the equations, so the solution will not be unique. To make the solution unique, $m-n$ forces, where m is the number of cables and n is the number of DOFs, can be specified in advance, and the problem solved as before. Alternatively, the problem can be modified and solved as a constrained optimization, wherein (17) is the constraint, and the cost function is arbitrary.

IV. Sag-Induced Cable Flexibility

Because cable-driven robotic manipulators are prone to vibration, determining their stiffness can be important for problems of design, analysis, and control. For many manipulators, the axial stiffness (elastic modulus) of the cables is the only source of flexibility that must be modeled, but for manipulators where cable sag is significant, a second type of flexibility, corresponding to the gravitational potential energy stored in the cable, must also be considered.

Sag-induced flexibility can be understood conceptually by considering a cable that is inextensible but has mass. Fixing one end of the cable and applying a force to the free end results in the cable being stretched out by the force, but not totally straightened. The cable will settle into a state of static equilibrium. In order to move the endpoint from equilibrium, the force must be increased or decreased. The relationship between the change in the force and the position of the endpoint is the stiffness of the cable, and this type of stiffness is clearly different from the axial stiffness of the cable.

Sag-induced stiffness is a 2-D quantity. The 2-D linear stiffness matrix $\mathbf{K}$ is defined as a relationship between force and displacement

$$\begin{bmatrix} dF_x \\ dF_z \end{bmatrix} = \mathbf{K}\begin{bmatrix} dx_{\mathrm{end}} \\ dz_{\mathrm{end}} \end{bmatrix}. \tag{18}$$

[1]This is not necessarily true if the drawing points of the cables are not fixed; see, for example, [16].

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For this problem, both the force and the displacement are defined as differentials representing small changes in force or position from equilibrium.

The relationship between the applied force and the motion of the endpoint can be found by evaluating (8) and (9) at $s = L_0$ [as given by (12) and (13)]. An intermediate step required for computing the stiffness is to first compute the compliance, the inverse of stiffness

$$\boldsymbol{\mathcal{C}} = \mathbf{K}^{-1} = \begin{bmatrix} \mathcal{C}_{11} & \mathcal{C}_{12} \\ \mathcal{C}_{21} & \mathcal{C}_{22} \end{bmatrix} \tag{19}$$

where

$$\mathcal{C}_{11} = \frac{\partial x_{\text{end}}}{\partial F_x} = \frac{L_0}{EA_0} + \frac{1}{\rho g}\left[\text{sign}(F_x)\left(\sinh^{-1}\left(\frac{F_z}{F_x}\right) + -\sinh^{-1}\left(\frac{-F_{z_0}}{F_x}\right)\right) - \frac{F_z}{\|\mathbf{F}\|} - \frac{F_{z_0}}{\|\mathbf{F}_0\|}\right] \tag{20}$$

$$\mathcal{C}_{12} = \frac{\partial x_{\text{end}}}{\partial F_z} = \frac{1}{\rho g}\left(\frac{F_x}{\|\mathbf{F}\|} - \frac{F_x}{\|\mathbf{F}_0\|}\right) \tag{21}$$

$$\mathcal{C}_{21} = \frac{\partial z_{\text{end}}}{\partial F_x} = \frac{1}{\rho g}\left(\frac{F_x}{\|\mathbf{F}\|} - \frac{F_x}{\|\mathbf{F}_0\|}\right) \tag{22}$$

$$\mathcal{C}_{22} = \frac{\partial z_{\text{end}}}{\partial F_z} = \frac{L_0}{EA_0} + \frac{1}{\rho g}\left(\frac{F_z}{\|\mathbf{F}\|} + \frac{F_{z_0}}{\|\mathbf{F}_0\|}\right) \tag{23}$$

where the forces F_x, F_z, $\mathbf{F}$, F_{z_0}, and $\mathbf{F}_0$ are as defined in Fig. 2. Note that $F_{z_0} = \rho_0 g L_0 - F_z$. The stiffness matrix finally can be obtained by inverting the compliance matrix, and is given explicitly by

$$\mathbf{K} = \frac{1}{\mathcal{C}_{11}\mathcal{C}_{22} - \mathcal{C}_{12}\mathcal{C}_{21}} \begin{bmatrix} \mathcal{C}_{22} & -\mathcal{C}_{12} \\ -\mathcal{C}_{21} & \mathcal{C}_{11} \end{bmatrix}. \tag{24}$$

A. Stiffness of an Inextensible Cable

The expression for the cable stiffness given above includes the elasticity of the cable. From (20)–(23), we can make the observation that the compliance of the cable can be separated into a sag-induced portion and an axial compliance portion

$$C_{\text{total}} = C_{\text{sag}} + C_{\text{axial}} \tag{25}$$

where

$$C_{\text{axial}} = \frac{L_0}{EA_0} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{26}$$

Subtracting the axial compliance of the cable from the total compliance leaves only the compliance due to the sag. In Section V, we look closer at the relative significance of these quantities.

B. Experimental Results

In this section, the sag-induced stiffness of the system is investigated experimentally. For the most part, the experimental verification of the cable shape in Section II-B is sufficient to imply that the stiffness predictions will be accurate; however, further experimental results help to provide more insight into the practical issues of sag-induced cable stiffness.

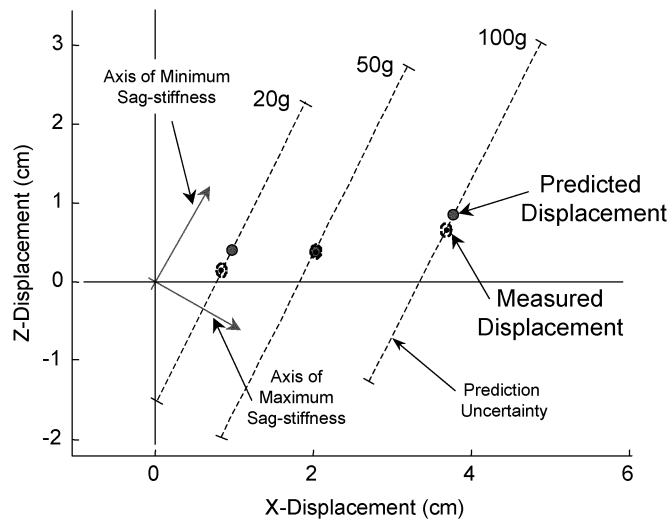

Fig. 5. Measured and predicted displacement corresponding to 20, 50, and 100 g weights.

The setup of this experiment is the same as for the cable-shape experiment in Section II-B. In this experiment, three precision weights with masses of 20, 50, and 100 g, respectively, were added to the original 623 g to increase the applied force and generate a displacement. The displacement and the change in the direction of the applied force corresponding to each added weight were captured using a digital camera and then measured from the images.

The results of this experiment are presented graphically in Fig. 5, which shows the average measured displacement of the three scenarios (each scenario consisting of three trials), the predicted displacement, and the uncertainty of the predicted displacement. The average measured displacements were all accurate to within 1 mm. The uncertainty in the prediction comes from the fact that the applied-force values (magnitude and direction) contained some measurement uncertainty. The direction measurement uncertainty had the greatest effect, even though the measurements were accurate to within 0.004 rad (0.23°). The predicted displacement, shown in the figure, is an average based on three independent measurements of the force vector. It should also be noted that there is some uncertainty in the direction perpendicular to the uncertainty shown, but it is negligible in comparison.

Combining the displacement observations in this section with the results from the cable-shape experiment in Section II-B, to within the precision of our experiments we can conclude that the stiffness model based on displacement is rather good. At the same time, however, the results bring up the question of how sensitive the model (and the system) is to changes in the applied force.

C. Displacement Sensitivity

The prediction uncertainty that arose in the previous experiment corresponds to the displacement sensitivity of the cable. Fig. 6 shows the predicted displacement sensitivity of the cable

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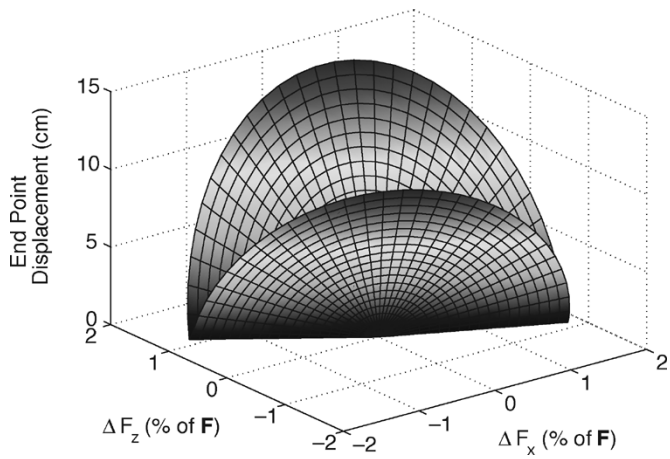

Fig. 6. Predicted displacement sensitivity of experimental cable.

from the experiment to changes in the applied force. The displacement is plotted over a uniform 2% change in the applied force in all directions. The nominal applied force is the same as that for the experimental system shown in Fig. 4, generated by the 623 g mass. The sensitivity corresponds closely to the compliance of the cable. As one might expect, the displacement sensitivity is greatest in roughly the direction of minimum stiffness (illustrated in Fig. 5). A small change in the force in this direction can yield a large change in the displacement.

Stiffness sensitivity is also significant. For a 2% change in the applied force, the maximum stiffness could increase by as much as 500% from the nominal value. As such, stiffness sensitivity can have a direct impact on the dynamics of the manipulator, and can affect the validity of the linearized stiffness.

V. Sample Manipulators

In this section, two examples, one planar and one spatial, are provided to show how the preceding cable-sag analysis can be synthesized for cable-driven mechanisms. The planar mechanism has three DOFs, and was chosen both for simplicity, and in order to clearly show the cable sag without the distortions of plotting in 3-D. The spatial manipulator, on the other hand, has six DOFs and is used to further illustrate interesting characteristics of cable sag.

A. Planar Manipulator

Fig. 7 shows a sample 3-DOF manipulator in a configuration where cable sag is clearly evident. The parameters used for this manipulator were chosen to roughly correspond to those for the proposed FAST system. The position and orientation of the end-effector is $(x, z, \phi) = (78.3\text{ m}, -133.6\text{ m}, -0.23\text{ rad})$, while the fixed parameters of the system are the linear cable density, $\rho_0 = 5.51$ kg/m, the mass and rotational inertia of the mobile platform, $m = 4000$ kg, and $I = 130\,000$ kg m^2, The distances from the center of mass of the end-effector to the ith cable attachment point, $b_1 = -10$ m, $b_2 = -8.8$ m, $b_3 = 10$ m, gravity $g = 9.8$ m/s^2, and the position of the cable base attachment points, $x_{01} = -250$ m, $x_{02} = -88.2$ m, $x_{03} = -250$ m, and $y_{01} = y_{02} = y_{03} = 0$. Each cable has an unstrained cross-sectional area of $A_0 = 7.1 \times 10^{-4}$ m^2 (3-cm diameter) and an

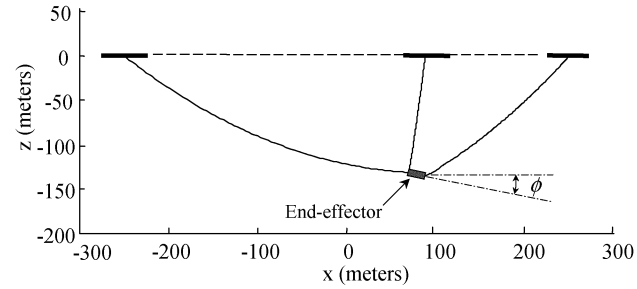

Fig. 7. Cable-driven planar mechanism.

elastic modulus of $E = 200$ GPa—this equates to an axial stiffness of approximately 570 kN/m for a 250–m-long cable.

1) Inverse Kinematics Initial Guess: In calculating the inverse kinematics, we found that a good initial guess was key to converging to the actual solution. Using a massless cable approximation to generate an initial guess worked well for most configurations we encountered. The advantage of this approach is that when the cables are massless, the inverse kinematics and the corresponding cable forces can be computed in closed form. For some cases, this initial-guess scheme was not successful. When the solution did not converge for the initial guess, we found that we could coerce convergence by starting with massless cables and iteratively solving the inverse kinematics as the density is increased until it reaches the desired level, each time using the previous solution as the next initial guess.

2) Inverse Kinematics Results: The inverse kinematics solution for this manipulator highlights the effect of the cable sag. The strained cable lengths were found to be, from left to right, 350.0, 132.9, and 212.2 m. These lengths differ from those of an ideal manipulator with massless cables by 5.4, 0.01, and 0.8 m, respectively. The significance of this calculation can be seen when the massless cable solution is used to position the actual manipulator; the resulting error in position of the end-effector would be $(\Delta x, \Delta z, \Delta\phi) = (-3.2\text{ m}, 2.3\text{ m}, 0.20\text{ rad})$.

3) Observations From the Inverse Kinematics: An unorthodox result that occasionally appeared when solving the inverse kinematics was that of excessive droop, where one or more of the cables hung from the end-effector, providing a downward force. While these results were all mathematically valid solutions, they would perhaps not be valid in a practical sense. These solutions are interesting, nonetheless, because they have the potential to make areas of the workspace accessible that would not be otherwise, were the cables massless.

4) Manipulator Stiffness: Multidimensional stiffness can be a rather troublesome characteristic to objectively analyze, particularly when both translational and rotational stiffness quantities coexist in the same stiffness matrix. For this reason, a more holistic measure, linearized natural frequency, will be used instead. Natural frequency is similar to stiffness in many regards, and has shown promise in design and analysis of other types of parallel kinematic mechanisms [17], [18].

To obtain the natural frequencies of the manipulator in a given configuration, the equations of motion must first be linearized. To obtain these linearized equations, it is not necessary to first fully derive the nonlinear equations of motion. The form of the resulting linear equations is known, so they can be obtained

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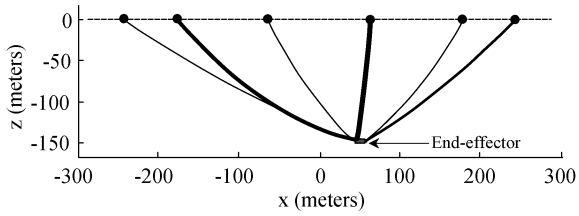

Fig. 8. 6-DOF mechanism.

using a more direct method, such as that presented in [19]. The equations of motion in linearized form (though not fully linearized) are

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = 0 \tag{27}$$

where

$$\mathbf{K} = \sum_{i=1}^{3} \mathbf{A}_i^T \mathbf{K}_i \mathbf{A}_i \tag{28}$$

$$\mathbf{A}_i = \begin{bmatrix} 1 & 0 & -a_i \sin\theta \\ 0 & 1 & a_i \cos\theta \end{bmatrix}. \tag{29}$$

Fully linearizing these equations of motion requires evaluating each $\mathbf{A}_i$ at the desired position and orientation of the manipulator: the equilibrium point. This can be confirmed with a Taylor expansion. Note, of course, that (27) is already written in a partially linearized and simplified form, and each $\mathbf{K}_i$ is implicitly variable, a function of both the position and orientation of the manipulator. The already linearized form of $\mathbf{K}_i$ is obtained directly from (24).

5) Results: With inextensible cables, the manipulator in Fig. 7 was calculated to have natural frequencies, from lowest to highest, of 0.79, 3.24, and 40.8 rad/s. With flexible cables, the natural frequencies become 0.79, 3.19, and 24.1 rad/s. The difference between the two cases is primarily evident for the highest frequency. This indicates that most of the manipulator compliance is due to the cable sag rather than the axial flexibility of the cables. While this is only one example, the result indicates that sag-induced flexibility can have a significant influence on the stiffness, and thus the behavior, of a large-scale cable-driven manipulator.

B. 6-DOF Manipulator

The second example is the 6-DOF manipulator shown in Fig. 8. This manipulator is based on a preliminary design of the FAST manipulator described in Section I-A. The drawing points for its cables are evenly spaced on a circle with a radius of 250 m in the $z = 0$ horizontal plane. The origin of the coordinate system lies at the center of the circle. The end-effector is disk-shaped, with a radius of 8 m and negligible height. The mass of the end-effector is 10 000 kg. The cables have the same properties as those in Section V-A: $\rho_0 = 5.51$ kg/m, $E = 200$ GPa, and $A_0 = 7.1 \times 10^{-4}$ m^2.

1) Inverse Kinematics Results: For this inverse kinematics example, the center of mass of the end-effector is located at $(x, y, z) = (50, 50, -150)$. The orientation is such that the end-effector is pitched by 0.2 rad (11.5°) about the x-axis.

TABLE II
NATURAL FREQUENCIES OF SPATIAL MANIPULATOR

Mode	Natural Frequencies (rad/s)		
	Inextensible	E = 200 GPa	E = 20 GPa
1	0.79	0.79	0.78
2	0.89	0.89	0.87
3	1.70	1.70	1.65
4	2.77	2.76	2.67
5	3.46	3.44	3.28
6	4.52	4.49	4.23

With inextensible cables, the inverse kinematics yields cable lengths of 264.56, 324.09, 351.37, 322.05, 264.97, and 229.96 m. For elastic cables, with properties as listed above, the strained cable lengths that solve the inverse kinematics are nearly identical to the inextensible cable lengths, varying by no more than 0.1 m or 0.037%, while the corresponding cable strain is less than 0.27 m for all cables. Decreasing the elastic modulus by a factor of ten yields a maximum difference between the inextensible cable solution and the strained cable solution of less than 1 m or 0.37%, in which case, the strain in the cables is, at most, 2.60 m.

2) Manipulator Stiffness: Again, the linearized natural frequencies of the manipulator are used to indirectly investigate its stiffness properties. Here we assume that the end-effector has a rotational inertia about all axes of $I_{ee} = 33.3 \times 10^3$ kg m^2. The natural frequencies were computed for three different values of cable elasticity (including the inelastic case), and the results are listed in Table II.

The results for this 6-DOF system are pretty consistent with the results found for the planar system. The difference between the different levels of cable stiffness is again primarily evident for the highest frequencies (and just slightly at that, varying at the maximum frequency by only about 6%). Again, this indicates that most of the manipulator compliance is due to the cable sag rather than the axial flexibility of the cables.

An obvious question to ask is, at what point does the cable elasticity become significant? The answer to this is dependent on the manipulator, but if the cables in our example had an elastic modulus of 1/100 of the original, the maximum frequency would still only decrease by 34%, while the minimum frequency would decrease by only 7.6%. These results further indicate that cable sag can be a significant source of manipulator flexibility.

VI. CONCLUSION

This paper addressed several important issues related to the statics of cable-driven manipulators. Solutions to the problems of inverse kinematics and cable-sag-induced flexibility were presented. The analysis of the sample systems in Section V illustrated the synthesis of the methods developed in the previous sections, and showed that cable sag can indeed have a large effect on both the inverse kinematics and the stiffness of a cable-driven manipulator. This research can readily be applied to other manipulators for which cable sag is significant to better analyze those manipulators. Furthermore, this research can potentially be extended to address other important issues, such as adjusting workspace calculations to account for cable sag and incorporating cable-sag predictions into the manipulator design process.

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